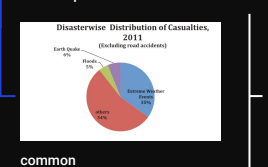
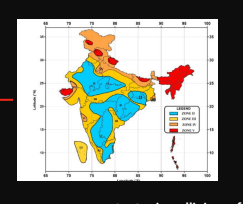
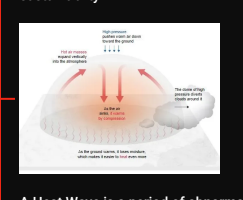
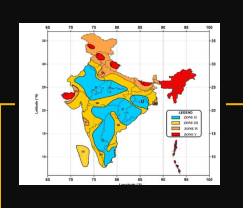
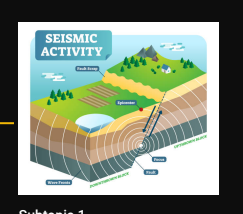
Diaster

- Main Topic 1
- Diaster
- Framework
  - Intro
    - Definition by instn.
    - Data
      - recent event
      - % under prone
  - Body
    - diagram
    - causes
    - impacts
      - social
      - economic
      - environmental
    - challenges
    - Measure
      - adaptation
      - mitigation
    - Govt programme
      - National level
      - International level
  - conclusion
    - forward looking
    - alining with gloabal sdg or framework or programme
- Main Topic 3
  - Framework
    - Intro
      - Definition by instn.
      - Data
        - recent event
        - % under prone
        - 
          common
          — India diaster report, 2011
    - Body
      - diagram
      - causes
      - impacts
        - social
          - High mortality (e.g., Malin 2014, Kedarnath 2013); destruction of settlements and heritage structures
          - Disruption of critical infrastructure (roads, bridges, power lines) — the 2021 Chamoli disaster damaged two hydropower plants causing over $200 million in losses
        - economic
          - Disruption of tourism and pilgrimage routes (e.g., Char Dham)
          - Loss of life leading to loss of productivity; massive compensation liabilities; damage to plant property and raw materials; disruption of supply chains
        - envtl
          - loss of biodiversity
          - Contamination of air, soil, and water bodies (surface and groundwater); adverse impact on flora and fauna; long-term ecological degradation
      - challenges
      - Measure
        - adaptation
          - Hazard Zonation: Mapping of GLOF Hazard Zones and restricting settlements/infrastructure in high-risk areas (Land Use Planning)
          - Early Warning Systems (EWS):
        - mitigation
      - Govt programme
        - National level — National Policy on Disaster Management;
        - International level
    - conclusion
      - forward looking
        - Technological Integration: Utilizing Synthetic-Aperture Radar (SAR) satellite imagery for all-weather monitoring and AI to detect small changes in lake areas
        - Transboundary Collaboration: Strengthening data-sharing mechanisms with upstream countries (China, Nepal) regarding river levels and glacial lake status
        - Community Focus: Strengthening Community-Based Disaster Preparedness (CBDP) to empower locals as first responders
      - alining with gloabal sdg or framework or programme — Align with SDG 13 (Climate Action) by strengthening resilience to climate-related hazards and SDG 11 (Sustainable Cities and Communities) by reducing the number of deaths and economic losses caused by disasters
  - Parlance — moraine-dammed glacial lakes
- Diaster Management
  - EQ — int
  - Drought
  - Landslide
  - GLOF
  - Industrial diaster
- Main Topic 4
  - Diaster Management
    - Intro — India has shifted from Relief Centric to Risk Reduction Paradigm
    - Data
      - India's rank in Climate Change Vulnearblity — 6th — GermanWatch
      - Area prone to diaster — NDMA
        - 70%
        - Flood — 12%
        - Landslide — 12%
        - EQ (III-V) — 60%
        - Drought — 68%
      - Costs
        - Economic
          - Climate Change — 2L Cr — NIDM
          - Air Pollution — 3% gdp — World Bank
          - Water Pollution — 6600 cr
          - Eq — major EQ can wipe out years of economic gains
          - Flood — 7K-10K Cr — RBI
          - Landslide — 1.5K - 2KCr — NDMA
          - Heatwaves — -
        - Lives
          - Air Pollution — 16L
          - water — 2L
          - Eq — 20K bhuj
          - Flood
            - 1600
            - People displace — 70-80L
          - Landslide — 500
          - Heatwaves — 150+ lives lost
        - GDP loss due to diaster — 2% — NIDM
        - People displace due to diaster in India b/w 2008-22 — 7Cr — IDMC (Intl. Displacement Monitoring Center)
      - Lack of following of Building codes
    - Body
      - Legislative Framework
        - DMA, 2005
          - NDMA — Under PM
          - NDRF — MHA
          - SDRF — States
          - NDMA Guidelines
        - Sendai for Diaster Risk Reduction — 2015-30
        - INSAT/ NISAR satellite — forecasting , monitoring and response
      - Govt recent steps
        - Sachet Apps for Cyclone Alerts — IMD
        - Apada Mitra — Trained village community voluniteers in 350+ districts
        - Vayulink (IAF) — For radio communication in NE
        - _X wathc app
    - Possible Diaster related questions
      - Landslides
        - Intro — Most common diaster in Himalayas , Western Ghats region causing human, infrastructural and ecological loss
        - Data
          - % area landslide prone — 12 — GSI , 2024
          - Examples
            - Waynad
            - Himachal
            - Shirur
        - Causes
          - Natural — Earthquake , GLOF , fragile ecology , rainfall
          - Manmade
            - mining and quarrying — Idduki and Waynad
            - unplanned hill cutting — Chardham
            - lack of Slope stability checks — Mid slope hotels and hydropower dams
        - Conclusion
          - Landslides are not just geological, but governance challenges — integrating science, community, and policy is the only sustainable path.
          - Mountain Criers Before it falls
      - Earthquake
        - Intro — Eq are among the most devastating sudden onset diaster exposing fault lines in diaster governance and infrastructure resilience.
        - Data
          - Area of India under siesmic zone III to V — 60%
          - Examples
            - Nepal Bihar — 6.4 M
            - Assam — 5.6 M
          - 
        - causes
          - natural — tectnoic collisions, faults line and sudden release of energy , volcanic activity, landslide and avalanche
          - manmade — large infra (dams) , nuclear tests, mining drilling,
        - Body — Vulnearbility factors
          - Unplanned urbanisation — Shimla
          - lack of infra resilience — % of building in zone IV-V with seisim meassures — 95%
          - high population density in seismic zones — Delhi
          - Tectonic Location
      - Floods
        - Intro — Most frequent diaster in India causing human , infrastructal and ecology loss
        - Data
          - Losses due to Flood — 7000 Cr — NIDM estimate
          - People affected annually by monsoon floods — 70-80L — MHA
          - Lives lost annually — 1600
        - Body
          - Types
            - Flash — Chamoi
            - Urban — Bengaluru
            - coastal — odisha and wb
            - riverine — Brahmputra
            - GLOF — sikkim
          - Causes
            - Natural Causes
            - Human
              - Glof, excess monsson , cloud bursts, cyclones
              - encroachments of river bed(delhi), irregular dam flow (bhakra) , unplanned urbanisation (bengaluru) , siltation
          - Govt Stance
            - Flood watch app
            - FLOODS (2020) ISRO satteline flood mapping
          - Conclusion — India cannot control the monsoon, but it can control its impact — through planning, foresight, and floodplain governance.
      - Heatwaves
        - Intro
          - Heatwaves are becoming more frequent, intense and longer in India, posing threat to agriculture,public health and urban sustainabilty
          - 
          - A Heat Wave is a period of abnormally high temperatures, more than the normal maximum temperature that occurs during the summer season in the North-Western parts of India.
          - NDMA
            - temp >40 in plain , > 30 in hill — departure avg < 40 — 5-6 deg
            - avg > 45 — irrespective
        - Impact
          - Heat cramp, exhaustion , stroke
          - work hours, productivity, fatigue, dizziness, dehydration
          - women
            - irregualrity in mensutral cycles, baterial infection in urinary tracts
            - anxity , anger, restlessness + double burden
        - Data
          - Nuber of heatwaves days
            - 2023 — 98
            - 2024 — 170
          - Deaths [2024] — Official 150+
          - hottest year in last 122 years — 2024 — IMD
          - States with Heat Action Plans — 23/28 — PIB
        - Govt stance
          - Ahemadabad first city for HAP reduction in mortality — 30-40%
          - India Cooling action plan (2019)
            - sustainable cooling by 2038
            - reduce cooling by half by 2038
        - Way forward
          - HAP — Formed, adequate budget
          - social protection — cool public space, staggered works hours, water accessibility
          - Infra
            - Green roof, vertical parks, green cover
            - Susutainable cooling
          - Community — SHG, PRI to distribute ORS
        - Conclusion — Heatwaves are no longer future threats — they are daily emergencies. The time to invest in cooling justice and resilience is now.
- EQ
  - Intro
    - Definition by instn. — An earthquake, also called a quake, tremor, or temblor, is the shaking of the Earth's surface resulting from a sudden release of energy in the lithosphere that creates seismic waves.
    - Data
      - recent event
        - A Magnitude 7.7 earthquake occurred along the Sagaing Fault, which separates the Indian Plate from the Eurasian Plate
        - Ongoing Crisis (2023-2025): Joshimath Land Subsidence, Uttarakhand. Situated in Seismic Zone V, the town faces severe structural instability due to its location on active fault lines (Vaikrita Thrust),
      - % under prone — Approximately 59% (specifically 58.6%) of India's landmass is liable to seismic hazard damage [NDMA]
        - Zone V (Very High Damage Risk - Intensity IX+): Covers ~11%
        - Zone IV (High Damage Risk - Intensity VIII): Covers ~18% of land area, including the Indo-Gangetic plains and Delh
        - Zone III (Moderate Damage Risk - Intensity VII): Covers ~30% of land area
  - Body
    -  
      Subtopic 1
    - causes
      - It results from the release of accumulated stress of the moving lithospheric or crustal plates [NDMA].
      - Natural
        - Tectonic Plate Movement — The primary cause is the continuous movement of the Indian Plate pushing into the Eurasian Plate at a rate of approximately 47-50 mm/year, leading to strain accumulation along the Himalayan arch — Presence of active faults such as the Main Central Thrust (MCT) and Vaikrita Thrust in the Himalayas (e.g., near Joshimath)
        - volcanic EQ — when movemetn of magma changes stress beneath earth surface and results in Eq — Many of the famous and most well known volcanoes are of this type, including those of the Ring of Fire
        - Landslides & Collapse: Large rockfalls or cave collapses. — 2011 Tohoku Earthquake, Japan:
      - Manmade
        - Reservoir Induced Seismicity (RIS)
          - triggered by the weight of massive water reservoirs.
          - The Koyna Earthquake (1967) in Maharashtra is a classic Indian example of RIS
        - Mining Activities: — heavy mining utilizing blasting techniques can degrade soil stability and trigger tremors — increased seismicity near major coalfields (like Korba in Chhattisgarh) and fracking sites
        - Underground nuclear explosions
        - Human-Induced/Developmental Projects: Construction of heavy infrastructure (hydroelectric projects, roads) on fragile mountain terrain, disturbing the geological equilibrium (e.g., Chamoli district issues)
    - impacts
      - social
        - During the period 1990 to 2006, more than 23,000 lives were lost due to 6 major earthquakes in India [NDMA]. — High Mortality: Over 95% of lives lost in past Indian earthquakes were due to the collapse of non-engineered buildings
        - Housing Loss: Collapse of residential structures, particularly non-engineered masonry buildings in rural and semi-urban areas
        - Displacement: Large-scale homelessness post-disaster; e.g., the Bhuj earthquake (2001) affected over 7,900 villages
        - Health Infrastructure disruption: Damage to hospitals and loss of medical personnel hampers emergency response
        - Vulnerable Groups: Disproportionate impact on women, children, and the elderly due to structural collapses during indoor hours
      - economic
        - Economic losses from major earthquakes can be estimated at ~1% of India's GDP
        - Total loss of public and private property in recent decades estimated at $4.97 billion
        - Damage to critical infrastructure like bridges, flyovers, power transmission, and water supply networks
        - Industrial Loss: Disruption to industrial clusters (e.g., Kutch 2001) affects supply chains and investment patterns
      - environmental
        - Liquefaction: Soil loses stiffness and behaves like a liquid during shaking, causing buildings to sink or tilt (common in Indo-Gangetic plains)
        - Landslides: — landslide in indonesia post eq
        - Change in River Courses: Intense shaking can alter landscape topography and divert river paths
    - challenges
    - Measure
      - adaptation
        - Seismic Retrofitting: Selective strengthening of critical lifeline structures (hospitals, schools, bridges) and high-priority buildings in Zones IV and V
        - Rapid Visual Screening (RVS): Implementation of RVS procedures to assess the vulnerability of existing building stock
        - Building Codes Compliance: Strict enforcement of IS 1893 (Criteria for Earthquake Resistant Design of Structures) and IS 13920 (Ductile Detailing)
        - Techno-Legal Regime: Modification of town planning bye-laws, Development Control Regulations (DCRs), and land use zoning to mandate earthquake-resistant features
        - Capacity Building: Training of rural artisans and masons in retrofitting and safe construction practices
        - Mock Drills: Conducting regular drills in schools, offices, and hospitals to test readiness
        - Zero Tolerance Policy: Adoption of a "Zero Tolerance" policy towards avoidable deaths by ensuring no new unsafe structures are built
      - mitigation — Mobile Apps: 'India Quake' for real-time parameter dissemination and 'Sagar Vani' for coastal alerts
    - Govt programme
      - National level
        - National Retrofitting Programme: Launched (2014) to systematically upgrade existing unsafe government structures
        - National Earthquake Risk Mitigation Project (NERMP): To strengthen structural and non-structural mitigation efforts in high-risk districts
        - National Building Code (NBC) 2016: Comprehensive guidelines regulating building construction activities across the country
      - International level
        - Coalition for Disaster Resilient Infrastructure (CDRI): Global partnership launched by India to promote resilient infrastructure systems [CDRI].
        - Sendai Framework: India is a signatory, committed to substantial reduction of disaster risk and losses by 2030 [UNDRR].
  - conclusion
    - forward looking
      - Tech Integration: Utilization of AI, Satellite Imagery, and GPS for monitoring crustal deformations and improving Early Warning Systems (EWS
      - Community Focus: Strengthening Community-Based Disaster Preparedness (CBDP) to empower locals as first responders
      - Strict implementation of the techno-legal framework to prevent the creation of new risks in the rapidly expanding built environment
    - alining with gloabal sdg or framework or programme — Aligns with SDG 11 (Sustainable Cities and Communities) by making human settlements inclusive, safe, resilient, and sustainable, and SDG 9 (Industry, Innovation, and Infrastructure) by building resilient infrastructure [UNDRR].
- Drought
  - Intro
    - Definition by instn.
      - Types
        - Meteorological Drought (significant decrease from normal precipitation, >10%), Hydrological Drought (depletion of surface and sub-surface water resources), and Agricultural Drought (inadequate soil moisture and rainfall to support healthy crop growth)
        - Ecological Drought :When the productivity of a natural ecosystem falls due to shortage of water and as a consequence of ecological distress, damages are induced in the ecosystem.
      - a situation where seasonal rainfall is less than 75% of the long-term average, indicating a deficit of over 25% — IMD
    - Data
      - recent event — Between 2020 and 2024, approximately two-thirds of the nation experienced drought — Drought in Numbers Report, 2022
      - % under prone
        - Approximately 68% of India's total cultivable area is vulnerable to drought — About 33% of the country's area is classified as having a chronic drought problem, while another 35% is drought-prone — Ministry of Agric and Cooperation
        - Arid/Semi-Arid Regions: Western Rajasthan (probability of drought once in 2 years) and Gujarat
        - Rain Shadow Areas: Interior Karnataka (e.g., South Interior Karnataka recorded the second-worst rainfall event since 1971 in 2011)
        - High Rainfall Zones: Even regions like Cherrapunji (Meghalaya), with >11,000 mm rainfall, face drought for almost nine months due to poor retention
  - Body
    - diagram — A flowchart illustrating the progression of drought types: Meteorological Drought (rainfall deficit) → Agricultural Drought (soil moisture stress, crop failure) → Hydrological Drought (reduced stream flow, reservoir depletion, groundwater decline) → Socio-Economic Drought (famine, migration, economic loss)
    - causes
      - Monsoon Failure/Variability: The Southwest monsoon accounts for 70-80% of annual rainfall; its failure or uneven distribution is the primary trigger
      - El Niño Effect: There is a clear association between El Niño events and weak monsoons in India, leading to widespread moisture stress
      - Over-exploitation of Groundwater: Rapid depletion of aquifers due to excessive pumping for irrigation, particularly in Punjab, Haryana, and Rajasthan
      - Deforestation & Degradation: Loss of forest cover reduces soil moisture retention and rainfall interception, aggravating drought conditions (e.g., Anantapur district, Andhra Pradesh)
      - Change in Land Use: Conversion of natural water bodies and wetlands into built-up areas reduces natural water storage and recharge capabilities
      - Inappropriate Cropping Patterns: Cultivation of water-intensive crops (like sugarcane and rice) in semi-arid and water-stressed regions (e.g., Maharashtra and Karnataka)
      - Lack of Water Conservation: Inadequate rainwater harvesting and neglect of traditional water storage structures (tanks, ponds) lead to poor recharge
      - Climate Change: Rising temperatures increase evaporation rates, turning meteorological droughts into agricultural droughts more quickly
    - impacts
      - social
        - Migration: Distress migration of small farmers and landless labourers to urban areas, leaving behind the elderly and children
        - Water Scarcity: Severe drinking water shortages for humans and livestock, leading to increased drudgery for women who fetch water
        - Health Risks: Malnutrition due to food insecurity, and spread of water-borne diseases due to the consumption of contaminated water
        - Suicides: Psychological trauma and debt traps lead to farmer suicides
        - Education Disruption: School dropout rates increase as children are engaged in labor or fetching water
        - Vicious cycle of unemployment,Poverty,Migration
      - economic
        - Crop Failure: Massive loss of Kharif and Rabi crops; e.g., during the 2002 drought, food grain production dipped significantly
        - Livestock Loss: Shortage of fodder and drinking water leads to distress sale or death of cattle, depleting assets of the poor
        - Power Generation: Reduced water levels in reservoirs impact hydroelectric power generation
        - Inflation: Shortage of essential commodities leads to price rises and increased need for food imports
        - Industrial Impact: Agro-based industries (sugar, cotton, textiles) suffer due to raw material shortages
        - The report also noted that between 1998 and 2017, severe droughts in India caused a 2 to 5% decline in the GDP — Drought in Numbers Report, 2022
      - environmental
        - Desertification: Extended drought accelerates land degradation and desertification, particularly in arid zones
        - Biodiversity Loss: Drying of wetlands and water bodies threatens flora and fauna
        - Groundwater Depletion: Increased reliance on aquifers causes long-term declines in water tables
    - challenges
      - Late Declaration: States often declare drought late (Dec/Jan), delaying relief measures which should start mid-season
      - Assessment Methodology: Variance in drought declaration criteria among states; some use 'district' as a unit while others use 'tehsil/mandal', leading to exclusion of affected pockets
      - Relief Centric Approach: The administrative focus often remains on post-disaster relief rather than pre-disaster mitigation and preparedness
      - Monitoring Gaps: Sparse network of ground-based observations (rain gauges, soil moisture sensors) hinders micro-level forecasting
    - Measure — adaptation
      - govt
        - Rainwater Harvesting: Mandatory implementation of rooftop harvesting and revival of traditional water bodies (tanks, johads)
        - Check Dams & Percolation Tanks: Construction of structures to arrest runoff and recharge groundwater in watershed areas
        - Inter-basin Transfer: Transfer of water from surplus to deficit basins (e.g., river interlinking projects where feasible).
        - Canal Maintenance: Lining of canals to reduce seepage losses and modernization of irrigation systems.
        - Afforestation: Planting drought-resistant and bio-diesel species (Jatropha, Pongamia) to improve soil moisture and provide alternative income
      - people
        - Crop Diversification: Promoting drought-resistant, short-duration, and less water-consuming crops (millets, pulses)
        - Rainwater harvesting : By making high bunds around the fields, adopting of terrace cultivation, and planting trees on the bunds of fields, the use of rainwater can be maximised.
        - Suitable farming methods for arid areas: Like Production of coarse and hardy cereals; conservation of soil moisture by deep ploughing, storing water in ponds and tanks and use of sprinklers for irrigation.
        - Micro-Irrigation: Expanding coverage of drip and sprinkler irrigation systems under schemes like PMKSY to improve water use efficiency
        - Crop Insurance: Robust implementation of PMFBY to provide financial safety nets against crop failure
        - Community Participation: Involving PRIs and Women Self-Help Groups (SHGs) in water budgeting and management
        - Drought Monitoring Cells (DMCs): Establishing DMCs at state levels to collate weather data and issue advisories
    - Govt programme
      - National level
        - Mahatma Gandhi National Rural Employment Guarantee Scheme (MGNREGS): Provides guaranteed employment; builds water conservation assets; allows increasing workdays during drought
        - Pradhan Mantri Krishi Sinchayee Yojana (PMKSY): Focuses on "Har Khet Ko Pani" (water for every farm) and "Per Drop More Crop" (micro-irrigation)
        - National Rainfed Area Authority (NRAA): Expert body to provide knowledge inputs for dryland and rainfed agriculture management
        - Integrated Watershed Management Programme (IWMP): Restores ecological balance by harnessing, conserving, and developing degraded natural resources
        - National Agricultural Drought Assessment and Monitoring System (NADAMS): Developed by ISRO/Department of Space for remote sensing-based drought assessment
        - NDMA Guidelines: Comprehensive guidelines issued in 2010 focusing on a paradigm shift from relief to mitigation
      - International level
        - International Cooperation: Following the Sendai Framework for Disaster Risk Reduction and UNCCD (United Nations Convention to Combat Desertification) goals
        - Integrated Drought Management Programme; UNCCD's Drought Resilience, Adaptation and Management Policy (DRAMP) Framework; Global Drought Information System etc.
  - conclusion
    - forward looking
      - Tech Integration: Utilization of Satellite Remote Sensing (NDVI, NDWI) and AI-based modeling for real-time drought monitoring and early warning
      - Establishing the India Drought Management Centre (IDMC) as an autonomous body to provide expert guidance
      - Integrating drought mitigation into developmental planning at all levels (District/State/National)
    - alining with gloabal sdg or framework or programme — Directly aligns with SDG 2 (Zero Hunger) by ensuring food security, SDG 6 (Clean Water and Sanitation) through efficient water management, and SDG 15 (Life on Land) by combating desertification
- Main Topic 8
- Landslide
  - Intro
    - Definition by instn.
      - A landslide is defined as "a movement of mass of rock, earth or debris down a slope" under the influence of gravity
      - Indian Institute of Technology Delhi has made the first high resolution landslide susceptibility map for India and the map data is available for free
      - Most common diaster in Himalayas , Western Ghats region causing human, infrastructural and ecological loss
    - types — rock falls, topples, debris flows, and creep
    - Data
      - recent event
        - Wayanad Landslides, Kerala (30 July 2024).
        - Sikkim GLOF/Landslide Cascade (October 2023/2024 context).
        - Chamoli — flash flood of 2021 in the Chamoli district of Uttarakhand which claimed at least 72 lives was due to an avalanche
        - Himachal
        - Shirur
      - % under prone — % Area Prone: Approximately 0.42 million sq. km, covering nearly 12.6% of India's land area (excluding snow-covered area), is prone to landslide hazards — ISRO's Landslide Atlas of India Findings:
        - Northwest Himalayas: Contributes 66.5% of landslides in India (J&K, Himachal Pradesh, Uttarakhand)
        - Northeast Himalayas: Accounts for 18.8% (Sikkim, Darjeeling, Seven Sisters)
        - Western Ghats: Accounts for 14.7% (Kerala, Karnataka, Goa, Maharashtra)
  - Body
    - diagram
    - causes
      - Rainfall (Natural): Rainfall is the primary natural triggering factor; variability in rainfall patterns and pore water pressure saturation destabilize slopes
      - Gravity: Gravity is the primary reason for the landslides. Steeper slopes coupled with gravitational force can trigger a massive landslide
      - Seismicity (Natural): Earthquake-induced landslides (co-seismic) occur due to ground shaking which activates existing slides or creates new ones (e.g., Sikkim 2011, Kashmir 2005)
      - Unplanned urbanization/encroachments, deforestation (removing root cohesion), road cutting, and blocking of natural drainage patterns significantly increase vulnerability
      - Natural — Earthquake , GLOF , fragile ecology , rainfall
      - Manmade
        - mining and quarrying — Vibrations from the blasts can degrade soils in the locations prone to landslides, such as in the Himalayas. — Idduki and Waynad
        - unplanned hill cutting — Chardham
        - lack of Slope stability checks — Mid slope hotels and hydropower dams
        - Urban encroachment into forests, disturbance of natural drainage, and cutting and deep excavations on hill slopes to develop buildings and infrastructure have contributed to a huge increase in landslides in Uttarakhand — Joshimath in 2023 was mainly due to a lack of infrastructure for drainage, an increase in human population, and rapid expansion of infrastructure to cater for the tourism potential.
        - Climate change: The intense and sudden rainfall due to climate change as well as the melting of the glaciers in the Himalayas have aggravated the problems of landslides in India
    - impacts
      - social
        - Global Rank: India is among the top 5 countries with the highest landslide risk, where the estimated loss of life per 100 sq. km per year is greater than one — ISRO Landslide Atlas of India
        - High mortality (e.g., Malin 2014, Kedarnath 2013); destruction of settlements and heritage structures
        - Disruption of critical infrastructure (roads, bridges, power lines) — the 2021 Chamoli disaster damaged two hydropower plants causing over $200 million in losses
      - economic
        - Globally, landslides contribute to an estimated average annual economic loss of $26 billion
        - Disruption of tourism and pilgrimage routes (e.g., Char Dham)
      - environmental — Formation of Landslide Dams (e.g., Parechhu river blockage) which pose flash flood risks upon breaching,. Loss of topsoil and biodiversity
    - challenges
      - Technical: Lack of high-resolution (1:10,000 scale) hazard maps for local planning
      - Difficulty in real-time monitoring and forecasting due to the localized nature of landslides
      - Administrative: Implementing and enforcing building codes and land-use regulations in difficult mountainous terrain
    - Measure
      - adaptation
        - Structural
          - Slope Stabilization: Use of retaining walls, soil nailing, rock bolting, and anchors
          - Drainage Correction: Construction of catch-water drains, contour drains, and sub-surface drainage to manage pore water pressure
          - Bio-engineering: Afforestation and use of geosynthetics (e.g., coir logs, jute) to arrest soil erosion Landslide Monitoring involves deploying sensors, satellite imagery, and ground-based instruments to detect ground movement and potential landslide precursors.
          - Landslide Monitoring involves deploying sensors, satellite imagery, and ground-based instruments to detect ground movement and potential landslide precursors. — Landslip Detection System: It was developed by the Indian Institute Of Technology-Mandi (IIT-Mandi), the Indian Army, and the Defence Research and Development Organisation (DRDO)
        - Non-structural
          - Land Use Zoning: Restricting development in high-hazard zones and strictly enforcing building codes
          - Early Warning Systems (LEWS): Developing rainfall threshold-based warning systems (e.g., I-D models)
      - mitigation
    - Govt programme
      - National level
        - National Landslide Risk Management Strategy (NLRMS) 2019: Addresses hazard mapping, monitoring, and capacity building — The NLRMS aligns with the Sendai Framework to enhance resilience, but effective implementation requires interstate coordination and local participation.
        - National Landslide Susceptibility Mapping (NLSM): Programme by Geological Survey of India (GSI) to map landslide susceptibility on a 1:50,000 scale
        - Landslide Atlas of India (2023): Released by ISRO/NRSC, providing a database of ~80,000 landslides for risk analysis
        - NDMA Guidelines (2009): Framework for management of landslides and snow avalanches
      - International level
        - Sendai Framework: Focuses on understanding disaster risk and investing in DRR for resilience
        - Coalition for Disaster Resilient Infrastructure (CDRI): Promotes resilient infrastructure in mountain regions
  - conclusion
    - forward looking
      - Development of mobile apps like 'Bhuvan FLIM' for field data collection
      - Training local communities (Naula/Dhara maintenance) and integrating indigenous knowledge for early warning and response
      - Generation of User-Friendly Landslide Hazard Maps — Utilizing InSAR (Interferometric Synthetic Aperture Radar), LiDAR, and UAVs (Drones) for precise slope monitoring and kinematics analysis
      - Following International Best Practices: Brazil's SNAKE System: The SNAKE System is the Landslide Early Warning System (LEWS),
    - alining with gloabal sdg or framework or programme — Aligns with SDG 11 (Sustainable Cities and Communities) by reducing the number of deaths and economic losses caused by disasters, and SDG 15 (Life on Land) by restoration of degraded land and soil
    - Conclusion
      - Landslides are not just geological, but governance challenges — integrating science, community, and policy is the only sustainable path.
      - Mountain Criers Before it falls
- Main Topic 11
- GLOF
  - Intro
    - Definition by instn.
      - GLOFs occur when a substantial amount of water stored in a glacial lake is suddenly discharged.
      - It involves the release of millions of cubic metres of water and debris in a short period, causing catastrophic flooding downstream
    - Data
      - recent event
        - The recent breaking off of a portion of the Nanda Devi glacier in Joshimath, Uttarakhand, resulted in significant flooding in parts of the state.
        - Event: Sikkim GLOF (October 2023 - Context for current mitigation). — Triggered by South Lhonak Lake burst; destroyed 31 major bridges, damaged the Chungthang Dam
        - Global Context: 2025 has been designated as the "International Year of Glaciers' Preservation" by the UN to address cryosphere risks
      - % under prone — Globally, 15 million people are exposed to impacts from potential GLOFs, with the highest exposure in the High Mountain Asia region
  - Body
    - diagram
    - causes
      - Fast Slope Migration: Rapid slope movement (slides, falls, avalanches) into the lake leads to displacement waves, causing the dam to overtop or triggering an immediate dam breach — For instance, 50% of GLOFs in the Hindu Kush Himalaya are triggered by ice avalanches — NDMA
      - Heavy Rainfall/Snowmelt and Cascading Processes: Intense rainfall or rapid snowmelt, along with cascading processes, result in a significant increase in water intake into the lake, potentially leading to a flood from an upstream lake.
      - Earthquake: Earthquakes can directly cause dam breaches and failures, resulting in a GLOF
      - Climate change
        - retreat of glaciers and formation of pro-glacial lakes
        - permafrost thaw
        - Black Carbon deposition reduces albedo, accelerating melt
    - impacts
      - social — Catastrophic loss of life and displacement in downstream communities. The events are sudden, often giving little lead time for evacuation
      - economic — GLOFs carry massive sediment loads that silt up reservoirs, reducing their economic life — Severe damage to Hydro-electric Projects (HEPs) (e.g., Chamoli 2021, Sikkim 2023)
      - environmental — Extreme channel erosion, sedimentation of riverbeds, and destruction of forests and arable land downstream
    - challenges
      - Monitoring Constraints: Inaccessibility of high-altitude terrain makes ground-based monitoring difficult; reliance on remote sensing often faces issues with cloud cover during monsoons
      - Short Lead Time: GLOF events can occur within minutes, challenging standard early warning dissemination protocols
    - Measure
      - adaptation
        - Lake Level Lowering: Pumping or siphoning water out of the lake to reduce hydrostatic pressure. Controlled breaching of moraine dams
        - Channel Construction: Construction of spillways or outlets to regulate water discharge (e.g., work done at South Lhonak Lake)
        - Check Dams: Construction of check dams and deflection walls downstream to arrest debris flow and reduce impact energy
        - Early Warning Systems (EWS): Installation of Automatic Water Level Recorders (AWLR) and sensor-based alerts
      - mitigation
    - Govt programme
      - National level
        - National GLOF Risk Mitigation Programme: Led by NDMA, this focuses on the risk inventory of 195 lakes, installation of automated weather stations, and remote sensing
        - NDMA Guidelines (2020): Comprehensive guidelines on the Management of Glacial Lake Outburst Floods, emphasizing hazard assessment and monitoring
        - Hydro-Meteorological Action Plan (HMAP): J&K initiative involving AWLRs and Automatic Rain Gauges for real-time data
      - International level
  - conclusion
    - forward looking
    - alining with gloabal sdg or framework or programme
- Main Topic 13
- industrial diaster
  - Intro
    - Definition by instn.
      - Chemical disasters are scenarios involving disease, disability, or death on a large scale among humans, animals, and plants due to toxins or disease caused by chemical agents. They may result from fire, explosion, toxic release, or poisoning
      - Genesis Event: The Bhopal Gas Disaster (December 1984), where the release of Methyl Isocyanate killed over 2,000 people immediately and affected thousands long-term, brought a paradigm shift in Chemical Disaster Management (CDM) in India
    - Data
      - recent event
        - HPCL Refinery, Vishakhapatnam (1997): Vapour cloud explosion
        - IPCL, Nagothane (1990): Gas Cracker Complex explosion
      - % under prone — MAH Units: There are approximately 1,666 Major Accident Hazard (MAH) units in India
  - Body
    - diagram
    - causes
      - These MAH units handle a wide range of chemicals (raw materials, products, wastes) with flammable, explosive, corrosive, and toxic properties
      - Large quantities of chemicals are stored or processed in industries often located in densely populated areas, enhancing vulnerability due to haphazard construction
      - Process and Safety System Failures: Including technical errors (design defects, corrosion, metal failure), human errors (neglecting safety instructions), and organizational errors (poor emergency planning)
      - Natural Calamities: Natural hazards triggering chemical accidents (Natech), e.g., the release of acrylonitrile at Kandla Port during the 2001 earthquake or damage to phosphoric acid containment during the 1999 Orissa Super Cyclone
      - Transportation Accidents: Incidents during the transport of hazardous chemicals (HAZCHEM) by road, rail, air, or pipelines (e.g., BLEVE - Boiling Liquid Expanding Vapour Explosion)
    - impacts
      - social — Large-scale mortality and morbidity; long-term health effects such as genetic disorders, neurological deficiencies, and psychological trauma (Post-Traumatic Stress Disorder)
      - economic — Loss of life leading to loss of productivity; massive compensation liabilities; damage to plant property and raw materials; disruption of supply chains
      - environmental — Contamination of air, soil, and water bodies (surface and groundwater); adverse impact on flora and fauna; long-term ecological degradation
    - challenges
      - Lack of emphasis on CDM (chemical diaster managemnet) functions; deficiencies in On-Site and Off-Site emergency plans; inadequate enforcement of regulations
      - Technical: Absence of national-level risk assessment criteria; use of obsolete technologies by Small and Medium Enterprises (SMEs); lack of standardized safety audit procedures
      - Legal: Lack of legislation on land-use policy (buffer zones) around industries; absence of national regulations specifically for medical emergency management in chemical disasters
      - Infrastructural: Inadequate medical infrastructure for handling mass chemical casualties (shortage of specific antidotes, trained paramedics)
    - Measure
      - adaptation
        - Land Use Planning: Strict implementation of land-use policies to prevent settlements near hazardous industries.
        - 2. Risk Assessment: Mandatory Hazard Analysis (HAZAN) and Hazard and Operability Studies (HAZOP) for identification of risks.
        - 3. Awareness Generation: Sensitizing the community and stakeholders about "DOs and DON'Ts" during chemical accidents.
        - 4. Capacity Building: Training of personnel, including drivers of HAZMAT vehicles, and strengthening of the factory inspectorates
        - Tech Integration: Implementation of GIS-based Emergency Planning and Response Systems in major industrial clusters
        - Development of a National Register on potentially toxic chemicals
        - Establishing a National Action Plan on Chemical Industrial Disaster Management (NAP-CIDM) and ensuring rigorous
      - mitigation
    - Govt programme
      - National level
        - Legal Framework: Manufacture, Storage and Import of Hazardous Chemicals (MSIHC) Rules, 1989
        - Crisis Groups: Establishment of Central (CCG), State (SCG), District (DCG), and Local Crisis Groups (LCG) for coordination
        - Schemes: "Industrial Pocket-wise Hazard Analysis" by MoEF to identify accident potential in districts
        - Institutions: Setting up of Emergency Response Centres (ERCs) and Poison Control Centres
      - International level — ILO Convention: Adherence to ILO Convention No. C 174 on prevention of major industrial accidents
  - conclusion
    - forward looking
    - alining with gloabal sdg or framework or programme
- Sendai
  - Intro — The Sendai Framework is a global, voluntary agreement adopted in 2015 to substantially reduce disaster risk and losses by shifting focus from disaster response to prevention, preparedness, and resilience.
  - Body
    - what is it
      - Adopted at the Third UN World Conference on Disaster Risk Reduction (2015, Sendai, Japan).
      - Successor to the Hyogo Framework for Action (2005–2015).
      - Led by UN Office for Disaster Risk Reduction (UNDRR).
      - Timeframe: 2015–2030.
    - objective — Prevent new disaster risks, reduce existing risks, and strengthen resilience.
    - 4 priorities
      - Understanding disaster risk (data, risk mapping).
      - Strengthening disaster risk governance.
      - Investing in disaster risk reduction (structural & non-structural).
      - Enhancing disaster preparedness for effective response & "Build Back Better".
    - 7 targets
      - Reduce disaster mortality, affected people, economic loss, and damage to critical infrastructure.
      - Increase DRR strategies, international cooperation, and early warning coverage.
    - India and Sendai
      - ~75% of India's districts are vulnerable to extreme events (floods, droughts, cyclones, earthquakes).
      - India is among the top 10 disaster-prone countries globally.
      - India loses ~2–3% of GDP annually due to disasters (World Bank estimates).
    - Issues and challenges in implementation
      - A. High Exposure Due to Geography & Demography — Long coastline, Himalayan belt, floodplains.
      - B. Rapid Urbanisation & Unplanned Growth
        - Expansion into floodplains, wetlands, seismic zones.
        - Example: Urban flooding in Chennai (2015) and Bengaluru due to poor planning.
      - C. Weak Local-Level Capacity — Disaster management plans exist but poorly implemented at district/ULB level.
      - D. Inadequate Risk-Informed Development — Infrastructure projects often ignore hazard mapping. — Example: Hydropower & road projects in Himalayas increasing landslide risks.
      - E. Climate Change Multiplying Risks — Data: Extreme rainfall events increased by ~75% since 1950.
    - impacts apart from usual ones
      - C. Poverty & Inequality — Disasters push vulnerable populations into poverty traps.
      - D. Disaster–Development Vicious Cycle — Rebuilding diverts funds from development. leading to slow porgress on sdg
      - Economic Risk: By 2100, India is projected to lose around 3-10% of its GDP annually due to climate change impacts
      - Sea Level Rise: Millions in coastal cities like Mumbai, Kolkata, and Chennai face risks from sea-level rise, flooding, and salinization
    - India preparedness
      - NDMA , 2005 — Features — Funds
        - National Disaster Response Fund NDRF — 100:0 (center state share) — supplements diaster relief when diaster is classified as severe
        - SDRF
          - Hilly and NE — 90:10
          - other — 75:25
          - usage — immediate relief for food, shelter, medical care and compensation
        - issues
          - delayed and underapproved funds under NDRF
            - kerala demand — 2200 Cr
            - kerala recieved in waynad — 260 Cr
          - SDRF — funds usage restricted to immediate relief but not reconstruction or livlihood restoration
          - Outdated compensation
            - life lost — 4L
            - fully damaged house — 1.2L
          - discretion of center in declaring diaster as severe in absense of definition
          - aid release is procedural not automatic
          - weak allocation crieteria of FC ignoring local vulnerability and adaptability , exposure, etc.
        - Solution
          - Phillipines model of quick response funds through rainfall and fatality indices
          - FC
            - update relief norm cost,
            - revise allocation — comprehensive vulnerability index
            - disater assistance remaind grant based not debt based
          - procedural chairty to rules based partnership
      - National Disaster Management Authority (NDMA) — → Policy formulation, guidelines aligned with Sendai priorities.
      - National Disaster Management Plan (2016, updated) — → India's roadmap aligned with Sendai Framework.
      - Early Warning Systems — → IMD-led cyclone & weather warnings; success in Odisha cyclone evacuations.
      - Coalition for Disaster Resilient Infrastructure (CDRI) — → India-led global initiative promoting resilient infrastructure.
      - National Cyclone Risk Mitigation Project (NCRMP) — → Strengthens coastal resilience.
    - way ahead
      - Mainstream DRR into all development planning.
      - Strengthen urban resilience & risk-sensitive land-use planning.
      - Invest in local-level capacity & community preparedness.
      - Expand multi-hazard early warning coverage to last-mile.
      - Integrate climate adaptation with DRR.
      - Improve data sharing, risk modelling, and monitoring.
  - Conclusion
    - Effective implementation of the Sendai Framework through risk-informed development, strong local institutions, and climate-resilient planning is crucial to break India's disaster–development cycle and safeguard long-term growth.
    - It aligns closely with the Sustainable Development Goals (SDGs), particularly SDG 11 (Sustainable Cities and Communities), aiming to substantially reduce disaster risk and losses in lives, livelihoods, and health

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